Ch. 4 CW Pt. 2

Principles of Microeconomics

September 8, 2025

Instructions

Please answer on a separate sheet of paper and submit on Gradescope by the end of class.

1. Consider the following events: Scientists reveal that eating oranges decreases the risk of diabetes, and at the same time, farmers use a new fertilizer that makes orange trees produce more oranges. Please illustrate what effect these changes have on the equilibrium price and quantity of oranges.

2. Because bagels and cream cheese are often eaten together, they are complements.

   (a) We observe that both the equilibrium price of cream cheese and the equilibrium quantity of bagels have risen. What could be responsible for this pattern – a fall in the price of flour or a fall in the price of milk? Please illustrate and explain your answer.

   (b) Suppose instead that the equilibrium price of cream cheese has risen but the equilibrium quantity of bagels has fallen. What could be responsible for this pattern – a rise in the price of flour or a rise in the price of milk? Please illustrate and explain your answer.

3. Suppose that the price of baseball tickets at your future college is determined by market forces. Currently the demand and supply schedules are as follows:

Price	**Quantity Demanded**	**Quantity Supplied**
\$4	10,000 tickets	8,000 tickets
8	8,000	8,000
12	6,000	8,000
16	4,000	8,000
20	2,000	8,000

   (a) What is unusual about this supply curve? Why might this be the case?

(b) What are the equilibrium price and quantity of tickets?

(c) Your future college plans to increase total enrollment next year by 5,000 students. The additional students will have the following demand schedule:

Price	**Quantity Demanded**
\$4	4,000 tickets
8	3,000
12	2,000
16	1,000
20	0

What will be the new equilibrium price and quantity?

4. Assume that demand for coffee is given by $Q_D = 30 - 2P$, and supply is given by $Q_S = 4P$.

   (a) What is the equilibrium price of coffee?

   (b) What is the equilibrium quantity of coffee?

5. Suppose that global demand for kombucha is $Q_D = 144 - 12P$, and global supply is $Q_S = 16 + 4P$.

   (a) What are the equilbrium price and quantity in the global kombucha market?

   (b) Suppose domestic (i.e. not foreign) demand for kombucha is $Q_{D,Dom} = 80 - 8P$, and suppose foreign demand increases by 50%. What is the new equilibrium price in the global kombucha market?